DO NOW — RECALL

Trees and binary search trees: implementation

Name: _____ Class: _____ Date: _____

1. What does a null Next mean in a singly linked list?

2. In a BST, where does a smaller key go?

3. What is the base case of a recursive insertion reaching an empty child?

4. Insert 10,20,30 in that order into an empty unbalanced BST. What shape results?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

It marks the end of the list.

2. Model answer

Into the left subtree.

3. Model answer

Create and return a new node at that null position.

4. Model answer

A right-leaning chain 10→20→30.

Vocabulary — know these terms

binary search tree — Binary tree in which lower keys occur to the left and higher to the right.

root — Topmost tree node.

leaf — A node with no children.

in-order traversal — Visit left subtree, node, then right subtree.

successor — Next key in in-order sorted sequence.

NEW LEARNING

Read, annotate and explain

Find the starting node

The root is 40, so compare 25 with 40 and move left.

Repeat comparisons

25>20 goes right to 30; 25<30 goes left.

Insert at a null child

Create node 25 at the empty left-child position of 30. This implementation ignores duplicate keys: when key equals node.Key, neither subtree changes.

Explain one step in your own words, using the correct technical terms.

I DO — WORKED EXAMPLE

Follow the worked method

Insert 12 into the BST 20 (root), 10 (left), 30 (right).

Predict first

Compare 12 with 20, then 10.

Model the reasoning

Insert 12 as the right child of 10, preserving the BST property.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Insert 12 into BST root 15, left 9 and right 20. Identify its parent and position.

Question 2

Insert 50,30,70,20,40,35 into an empty BST. State parent and side of 35, and the in-order visit sequence.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

In BST 40 with children 20 and 60, and 20 with children 10 and 30, delete leaf 10. Explain a second case in which deletion is more involved.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

In-order traversal of a BST with distinct keys visits values in ascending order.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

Next step: revisit the model or a relevant revision-guide section if needed.